

GenesisGeo: Technical Report

Minfeng Zhu¹, Zi Wang¹, Sizhe Ji¹, Zhengtong Du¹, Shengqiang Tai², Junming Ke³, Xiao Deng³, Zanlang Yin⁴, Xiuqi Huang¹, Heyu Wang⁵, and Wei Chen¹

¹State Key Lab of CAD&CG, Zhejiang University

²Polytechnic Institute, Zhejiang University

³Hangzhou Research Institute of AI and Holographic Technology

⁴Volkswagen Group Innovation

⁵School of Mathematical Science, Zhejiang University

{minfeng_zhu, zi_wang, sizhe_ji, zhengtongdu, shengqiangtai,
huangxiuqi, wangheyu, chenvis}@zju.edu.cn

{junmingke, xiaodeng}@zjuqx.com, zanlang.yin@gmail.com

Abstract

Recent neuro-symbolic geometry theorem provers have made significant progress on Euclidean problems by coupling neural guidance with symbolic verification. However, most existing systems operate almost exclusively in a symbolic space, leaving diagram-based intuition largely unused during reasoning. For humans, geometric diagrams provide essential heuristics for identifying non-trivial auxiliary constructions. Meanwhile, visual language models (VLMs) still struggle with geometry due to the lack of high-quality data with geometric diagrams and reasoning supervision. In this paper, we introduce GENESISGEO-1M, a large-scale synthetic dataset for visual geometric reasoning that contains 1M multimodal geometry problems paired with machine-checkable proof traces. Building on this dataset, we formulate geometric learning as a multi-task training paradigm that jointly optimizes text-based proof generation and diagram-grounded proof generation, encouraging models to learn visual grounding and symbolic deduction. Extensive experiments show that our GENESISGEO-2B model achieves gold-medal-level performance on Olympiad geometry benchmarks, solving 29/30 problems on IMO-30, 63/95 on IMO-95, and 278/409 on HAGEO-409.

1 Introduction

Automated theorem proving in Euclidean geometry has made remarkable progress in recent years [Trinh et al. \(2024\)](#); [Duan et al. \(2025\)](#); [Chen et al. \(2025\)](#). Neuro-symbolic systems such as AlphaGeometry2 can reach gold-medal-level performance [Chervonyi et al. \(2025\)](#). These systems show that combining language-guided auxiliary construction search with symbolic deduction can support long-horizon geometric proofs. However, most existing methods solve problems primarily in a textual space. They encode the problem in a formal language, guide auxiliary construction search using language- or heuristic-based priors, and verify conclusions via symbolic deduction. This workflow overlooks visual cues (e.g., symmetry, alignment, concurrency, and cyclic patterns) that often suggest the correct auxiliary construction in Olympiad geometry.

In human problem-solving, diagrams serve as a perceptual workspace for forming hypotheses, checking constraints, and identifying useful auxiliary constructions. Experienced mathematicians often use visual cues to choose auxiliary constructions before writing a proof. This visual intuition is central to Olympiad geometry, where a solution often depends on adding the right auxiliary point, line, or circle. In contrast, current VLMs remain unreliable on geometry problems with diagrams [Pan et al.](#)

(2025); Lu et al. (2024). One reason is the lack of large-scale multimodal geometry data that pairs diagrams with machine-checkable proof traces. Existing multimodal datasets rarely expose models to the full reasoning pipeline required for Olympiad geometry, from perceptual pattern recognition to auxiliary construction and formal verification. This disconnect prevents models from learning how visual evidence informs symbolic planning in long-horizon proofs. As a result, current systems often fail to discover non-obvious auxiliary constructions that are critical for solving competition-level problems.

To address this gap, we introduce GENESISGEO-1M, a large-scale synthetic dataset for visual geometric reasoning with verified proof supervision. We enable this scale by reimplementing the symbolic deduction engine in C++ and equipping it with a carefully curated rule set (e.g., the *secant theorem*) that improves both deductive capability and runtime efficiency. Building on this engine, we develop an automated synthesis pipeline that generates challenging Olympiad-style problems. The pipeline samples construction-based problems, performs symbolic reasoning to obtain a formally checkable proof trace with the necessary auxiliary constructions, and renders the geometry constructions into a diagram with deterministic grounding of symbolic objects in the visual space. Each example includes a problem statement, a diagram, and a step-by-step symbolic proof trace. This design tightly aligns visual representations, symbolic premises, auxiliary constructions, and proof steps, enabling models to learn how perceptual cues inform formal deduction. Compared to prior datasets Gao et al. (2025); Wu et al. (2025), GENESISGEO-1M provides large-scale multimodal geometry problems that enable long-horizon, diagram-grounded geometric reasoning.

In addition, we introduce GENESISGEO-2B, a multimodal geometry model trained with a multi-task learning strategy, including text-based proof generation, and diagram-grounded proof generation. These tasks share the same underlying geometry but provide complementary supervision signals: text-based proof generation strengthens symbolic rigor without relying on visual shortcuts, and diagram-grounded proof generation teaches the model to use visual evidence when proposing auxiliary constructions. We fine-tune strong vision-language backbones on GENESISGEO-1M under this multi-task training strategy. Experiments show consistent gains on established geometry benchmarks. In particular, GENESISGEO-2B solves 29/30 problems on IMO-30, 63/95 on IMO-95, and 278/409 on HAGeo-409, suggesting that large-scale multimodal supervision improves diagram understanding and symbolic reasoning.

In summary, our contributions are fourfold:

- We release GENESISGEO-1M, a multimodal geometry dataset with diagrams and machine-checkable proof traces for visual geometric reasoning.
- We propose a multi-task training strategy that jointly learns diagram understanding, text-based proof generation, and diagram-grounded proof generation.
- We conduct extensive experiments on established geometry benchmarks and demonstrate substantial gains after fine-tuning on GENESISGEO-1M.
- We provide a new solution to IMO 2008 Problem 6 that requires only two auxiliary points, simplifying AlphaGeometry-based solutions.

2 Related Work

2.1 Geometry Problem Solving

Automated geometric reasoning has a long history, progressing from algebraic elimination to symbolic proof systems and, more recently, neuro-symbolic approaches.

Algebraic and symbolic solvers. Early works on automated geometry translate geometric constraints into algebraic elimination, including Wu’s method and related mechanical theorem-proving systems Wu (1978); Chou (1988), as well as Gröbner-basis procedures Kapur (1986). These algebraic approaches can be computationally efficient, but the resulting proofs are often difficult for humans to interpret. Alternative lines of work aimed to produce human-readable proofs, notably the area method Chou et al. (1993) and deductive databases Chou et al. (2000). Recent work shows that hand-crafted symbolic heuristics can outperform classic methods on Olympiad-level benchmarks Duan et al. (2025). However, purely symbolic search can become inefficient when auxiliary constructions rapidly expand the search space.

Neuro-symbolic solvers. Recent approaches address search explosion by integrating deep neural networks for auxiliary construction proposal. AlphaGeometry couples a deductive database with an algebraic engine and trains a language model on synthetic textual corpora to suggest auxiliary points [Trinh et al. \(2024\)](#). Subsequent systems enhance the symbolic engine and data construction (e.g., TongGeometry [Zhang et al. \(2026\)](#), SeedProver [Chen et al. \(2025\)](#), and AlphaGeometry2 [Chervonyi et al. \(2025\)](#)), solving the full IMO-30 set and achieving high success rates over broader collections of IMO problems. For example, TongGeometry applies tree search to explore auxiliary constructions [Zhang et al. \(2026\)](#). AlphaGeometry2 extends the domain-specific language with dynamic point moves and linear relations, achieving an 84% success rate on IMO problems from 2000–2024 [Chervonyi et al. \(2025\)](#). Despite these advances, most approaches still operate on symbolic inputs, without exploiting rich visual cues. We aim to learn construction-relevant auxiliary proposals from visual diagram evidence.

2.2 Multimodal Geometric Reasoning

Multimodal models such as ChatGPT and Gemini perform well on general vision tasks but remain unreliable at parsing geometric diagrams and performing multi-step reasoning [Wang et al. \(2024\)](#). Specialized models aim to close this gap by improving either *understanding* or *reasoning*.

Understanding-oriented work improves the mapping from pixels to geometric primitives and relations. G-LLaVA [Gao et al. \(2025\)](#) synthesizes geometric vision-language data to provide MLLMs with fundamental geometric knowledge. GeoVLMath uses cross-modal rewards to better ground auxiliary-line descriptions in the diagram [Guo et al. \(2025\)](#). MathMathCoder-VL, trained with ImgCode-8.6M, leverages code as supervision for cross-modal alignment and understanding [Wang et al. \(2025\)](#).

Reasoning-oriented work introduces step-by-step supervision to encourage multi-step deduction. Datasets such as GeoThought and GPSM4K provide intermediate reasoning traces and improve performance under chain-of-thought style training [Shi et al. \(2025\)](#); [Jaiswal et al. \(2024\)](#). GeoGen highlights hallucination issues in direct multimodal reasoning and proposes neuro-symbolic integration with translation and verification components [Pan et al. \(2025\)](#).

In contrast to prior work, we aim to push multimodal geometric reasoning to IMO-level difficulty by leveraging a multi-task training strategy that learns geometric visual cues while a symbolic engine verifies logical soundness.

2.3 Geometry Datasets

Geometry datasets broadly fall into (1) textual datasets that teach symbolic deduction and (2) multimodal datasets that pair diagrams with questions and proof traces.

Textural datasets. Large synthetic corpora have been central to recent neuro-symbolic provers. AlphaGeometry trains language models on 100M problems to learn auxiliary-construction priors and guide symbolic search [Trinh et al. \(2024\)](#). Follow-up systems scale the generation pipeline and search procedure (e.g., TongGeometry [Zhang et al. \(2026\)](#) and GeoGen [Bak et al. \(2020\)](#)) to further improve coverage and robustness. These corpora are effective for geometric reasoning in symbolic space, but they omit diagrams and therefore cannot teach the perceptual cues that humans rely on when selecting constructions.

Multimodal datasets. Human-curated benchmarks such as Geometry3K and GeoQA provide diagrams paired with formal annotations or executable programs [Lu et al. \(2021\)](#); [Chen et al. \(2021\)](#). Further, UniGeo unifies calculation and proving tasks with proving and program sequences that shares the same format [Chen et al. \(2022\)](#). AuxSolidMath curates exam-style problems with paired diagrams and aligned text for auxiliary-line reasoning [Guo et al. \(2025\)](#). For scaling multimodal training, Geo170K supplies diagram-text pairs [Gao et al. \(2025\)](#) and GeoThought further provide detailed thinking processes during problem-solving [Shi et al. \(2025\)](#). Recently, researchers employ a neuro-symbolic framework for generating geometric reasoning data (e.g., NeSyGeo [Wu et al. \(2025\)](#), GeoGen [Pan et al. \(2025\)](#), TrustGeoGen [Fu et al. \(2025\)](#)).

Despite rapid progress, the field still lacks large-scale multimodal training data at IMO-level difficulty. We introduce GENESISGEO-1M dataset with diagram-text pairs and machine-checkable proof traces, which enables gold-medal-level performance on Olympiad geometry benchmarks.

3 Preliminary of Symbolic Engine

This section introduces our symbolic engine, which operates over two geometry-specific DSLs, and performs forward closure via a Deductive Database and Algebraic Reasoning (DDAR) procedure. We also apply several accelerations to support large-scale generation.

Geometry-specific languages. A symbolic geometry engine requires explicit formal languages for geometric constructions and predicate descriptions. The **construction language** defines geometric elements (e.g., points, lines, and circles) with commands such as `rectangle A B C D` and `circle O A B C`. The **predicate language** describes geometric facts over points, such as congruence (`cong O A O C`) and parallelism (`para A B C D`).

DDAR engine. DDAR combines a rule-based deductive database [Chou et al. \(2000\)](#) with an algebraic reasoning module. The DD component applies rules of the form $P_1, \dots, P_m \rightarrow P_{\text{new}}$ to derive new predicates, repeating rule matching until no new predicates can be inferred. The AR module maintains quantitative constraints (e.g., lengths and ratios) as linear equations and uses Gaussian elimination to derive additional equalities. Together, DDAR can compute a complete closure of deducible facts.

Engine enhancement. We enhance the symbolic engine to increase its reasoning capacity through refined deductive rules and an improved arithmetic equation solver.

Rule Reduction. The primary motivation for pruning the baseline rule set is to optimize computational efficiency during rule matching. In automated theorem proving, an excessive number of candidate rules at each step can exponentially inflate the search space. Therefore, we eliminate redundant rules that are compositional (i.e., derivable from other rules) or logically trivial (e.g., describing simple algebraic symmetries already implemented by the AR engine).

Rule Expansion. Conversely, the introduction of new rules (marked with * in Table 1) is driven by the need to enhance geometric expressivity. The baseline rule set treats some metric facts, such as equal distances, in a purely algebraic way. Our new constructive rules (e.g., Circumcenter Definition) serve as semantic bridges that convert these metric constraints into explicit geometric objects (e.g., circles). We also incorporate hierarchical reasoning shortcuts, such as promoting similarity to congruence when a unit scale factor is detected. These rules bypass redundant searches for elementary congruence conditions, such as SSS (Side-Side-Side) and SAS (Side-Angle-Side). Finally, axiomatic rules like Unique Intersection ensure rigorous handling of point-identity scenarios. Table 1 shows the full rule set used in our GenesisGeo engine.

Notation. We use the following notation in the deductive rules:

- $\odot(ABCD)$: Points A, B, C, D are cyclic.
- $\odot(O; A)$: Circle with center O and radius OA .
- $\overline{ABC}$: True if Points A, B, C are collinear.
- $\text{Mid}(M, AB)$: Point M is the midpoint of segment AB .
- $AB : CD$: The ratio of the length of segment AB to the length of segment CD .
- $A \Rightarrow B$: Means if A is true then B is true.
- $\mathcal{O}(ABC)$: Represents the winding order of points A, B, C , used to distinguish between direct and reflective symmetry.
- $\triangle ABC \sim \triangle DEF$: Two triangles are similar and have the same orientation.
- $\triangle ABC \overset{R}{\sim} \triangle DEF$: Two triangles are similar **as mirror images** (opposite orientation).
- $\triangle ABC \cong \triangle DEF$: Two triangles are congruent and have the same orientation.
- $\triangle ABC \overset{R}{\cong} \triangle DEF$: Two triangles are congruent **as mirror images** (opposite orientation).
- $\angle ABC = \angle DEF$: The directed angle from line AB to BC is congruent to the one from line DE to EF modulo π .
- $\wedge$: Logical AND, true only if all connected conditions are true.
- $\neg$: Logical NOT, reverses the truth value of a statement.

Table 1: The full set of deductive rules used in GenesisGeo. Newly introduced rules are marked with an asterisk (*)

Rule Name	Logical Implication
I. Fundamental Incidence, Lines & Parallelism	
Line Definition by Parallelism	$\overline{ABC} \Rightarrow AB \parallel BC$
Collinearity of Parallel Lines	$AB \parallel AC \Rightarrow \overline{ABC}$
* Line Intersection Axiom	$\overline{PAB} \wedge \overline{QAB} \wedge \overline{PCD} \wedge \overline{QCD} \wedge \neg \overline{ABD} \Rightarrow P = Q$
Pappus's Hexagon Theorem	$\overline{ABC} \wedge \overline{PQR} \wedge \overline{XAQ} \wedge \overline{XPB} \wedge \overline{YAR} \wedge \overline{YPC} \wedge \overline{ZBR} \wedge \overline{ZCQ} \Rightarrow \overline{XYZ}$
Intercept Theorem	$AB \parallel CD \wedge \overline{OAC} \wedge \overline{OBD} \Rightarrow OA:OC = OB:OD$
Converse of Intercept Theorem	$OA:OC = OB:OD \wedge \overline{OAC} \wedge \overline{OBD} \wedge \neg \overline{ABC} \Rightarrow AB \parallel CD$
Intercept Theorem (Trapezoid)	$AB \parallel CD \wedge \overline{MAD} \wedge \overline{NBC} \wedge MA:MD = NB:NC \Rightarrow MN \parallel AB$
Proportional Transversals Theorem	$AB \parallel CD \wedge \overline{MAD} \wedge \overline{NBC} \wedge MN \parallel AB \Rightarrow MA:MD = NB:NC$
II. Metric Relations & Circles	
Definition of Midpoint	$AM = BM \wedge \overline{MAB} \Rightarrow \text{Mid}(M, AB)$
Property of Midpoint	$\text{Mid}(M, AB) \Rightarrow \overline{MAB}$
Midpoint Ratio Property	$\text{Mid}(M, AB) \Rightarrow AM:AB = 1:2$
* Segment Ratio Property	$\overline{ABC} \wedge \overline{DEF} \wedge AB:AC = DE:DF \Rightarrow AB:BC = DE:EF$
* Radii of Circle	$A, B, C \in \odot(O; A) \Rightarrow OA = OB = OC$
* Definition of Circumcenter	$OA = OB = OC \Rightarrow A, B, C \in \odot(O; A)$
Concyclic Equidistance Property	$A, B, C \in \odot(O; A) \wedge \odot(ABCD) \Rightarrow OA = OD$
Equidistance Criterion	$OA = OB \wedge OC = OD \wedge \odot(ABCD) \wedge AB \nparallel CD \Rightarrow OA = OC$
Inscribed Angle Theorem	$\odot(ABPQ) \Rightarrow \angle APB = \angle AQB$
Converse of Inscribed Angle Theorem	$\angle APB = \angle AQB \wedge \neg \overline{ABP} \Rightarrow \odot(ABPQ)$
* Definition of Secant Line	$\overline{PAB} \wedge OA = OB \Rightarrow \overline{PAB}$ is a secant of $\odot(O; A)$
III. Triangle Properties	
Angle Bisector Theorem	$DB:DC = AB:AC \wedge \overline{DBC} \wedge \neg \overline{ABC} \Rightarrow \angle BAD = \angle DAC$
Converse of Angle Bisector Theorem	$\angle BAD = \angle DAC \wedge \overline{DBC} \wedge \neg \overline{ABC} \Rightarrow DB:DC = AB:AC$
Median to Hypotenuse Theorem	$AB \perp BC \wedge \text{Mid}(M, AC) \Rightarrow AM = BM$
Concurrency of Altitudes	$AB \perp CD \wedge AC \perp BD \Rightarrow AD \perp BC$
Concurrency of Angle Bisectors	$\angle BAX = \angle XAC \wedge \angle ABX = \angle XBC \wedge \neg \overline{ABC} \Rightarrow \angle BCX = \angle XCA$
IV. Similarity & Congruence	
Definition of Similarity	<i>Direct</i> $\triangle ABC \sim \triangle PQR \Rightarrow \angle ABC = \angle PQR \wedge AB:BC = PQ:QR$
	<i>Reflective</i> $\triangle ABC \stackrel{R}{\sim} \triangle PQR \Rightarrow \angle ABC = \angle PQR \wedge AB:BC = PQ:QR$
* Unit Ratio Similarity	<i>Direct</i> $\triangle ABC \sim \triangle PQR \wedge AB = PQ \Rightarrow \triangle ABC \cong \triangle PQR$
	<i>Reflective</i> $\triangle ABC \stackrel{R}{\sim} \triangle PQR \wedge AB = PQ \Rightarrow \triangle ABC \stackrel{R}{\cong} \triangle PQR$
* CPCTC	<i>Direct</i> $\triangle ABC \cong \triangle PQR \Rightarrow AB = PQ \wedge \triangle ABC \sim \triangle PQR$
	<i>Reflective</i> $\triangle ABC \stackrel{R}{\cong} \triangle PQR \Rightarrow AB = PQ \wedge \triangle ABC \stackrel{R}{\sim} \triangle PQR$
AA Similarity Criterion	<i>Direct</i> $\angle ABC = \angle PQR \wedge \angle ACB = \angle PQR \wedge \neg \overline{ABC} \wedge \odot(ABC) = \odot(PQR) \Rightarrow \triangle ABC \sim \triangle PQR$
	<i>Reflective</i> $\angle ABC = \angle PQR \wedge \angle ACB = \angle PQR \wedge \neg \overline{ABC} \wedge \odot(ABC) \neq \odot(PQR) \Rightarrow \triangle ABC \stackrel{R}{\sim} \triangle PQR$
SSS Similarity Criterion	<i>Direct</i> $AB:BC = PQ:QR \wedge AC:BC = PR:QR \wedge \neg \overline{ABC} \wedge \odot(ABC) = \odot(PQR) \Rightarrow \triangle ABC \sim \triangle PQR$
	<i>Reflective</i> $AB:BC = PQ:QR \wedge AC:BC = PR:QR \wedge \neg \overline{ABC} \wedge \odot(ABC) \neq \odot(PQR) \Rightarrow \triangle ABC \stackrel{R}{\sim} \triangle PQR$
SAS Similarity Criterion	<i>Direct</i> $AB:BC = PQ:QR \wedge \angle ABC = \angle PQR \wedge \neg \overline{ABC} \wedge \odot(ABC) = \odot(PQR) \Rightarrow \triangle ABC \sim \triangle PQR$
	<i>Reflective</i> $AB:BC = PQ:QR \wedge \angle ABC = \angle PQR \wedge \neg \overline{ABC} \wedge \odot(ABC) \neq \odot(PQR) \Rightarrow \triangle ABC \stackrel{R}{\sim} \triangle PQR$

Arithmetic Rules. While the deductive engine handles qualitative geometric logic, the Algebraic Reasoning (AR) engine provides a quantitative foundation by converting geometric constraints into canonical arithmetic forms. This section details the mapping logic in Table 2 and the computational optimizations employed to maintain efficiency within the joint deduction loop.

To bridge the gap between symbolic predicates and numerical constraints, we define a mapping $\mathcal{M}$ that translates high-level geometric relations into algebraic equations (Table 2).

- **Linear Constraints:** Predicates such as *coll* (collinearity) and *para* (parallelism) are linearized using the concept of **directed angles**. For example, an angle equality $\angle(l_1, l_2) = \angle(l_3, l_4)$ is transformed into:

$$\text{dir}(l_1) + \text{dir}(l_4) - \text{dir}(l_2) - \text{dir}(l_3) = 0 \pmod{\pi}$$

- **Metric and Complex Constraints:** Distance-based predicates (*cong*, *eqratio*) and complex relations (*secant*) are mapped to logarithmic equations or polynomial identities (e.g., Power of a Point Theorem).

Together, these enhancements keep the engine symbolic and efficient, while expanding its deductive closure to cover exponential constraints.

Table 2: Catalog of Arithmetic Equations

Predicate	Equation
coll A B C	$\angle(AB) = \angle(AC) = \angle(BC)$ $AB + BC = AC$ (if B lies on segment AC)
aconst A B C D r	$\angle(CD) - \angle(AB) = r$
cong A B C D	$AB = CD$
eqangle A B C D E F G H	$\log(AB) = \log(CD)$ $\angle(AB) + \angle(AD) = 2 \cdot \angle(BD)$ (if C equals to A)
eqratio A B C D E F G H	$\angle(AB) - \angle(CD) = \angle(EF) - \angle(GH)$ $AB \cdot GH = CD \cdot EF$ $\log(AB) - \log(CD) = \log(EF) - \log(GH)$
para A B C D	$\angle(AB) = \angle(CD)$
perp A B C D	$\angle(AB) - \angle(CD) = \pi/2$ $AB^2 + AD^2 = BD^2$ (if C equals to A)
rconst A B C D r	$AB = CD \cdot r$
secant O A B P	$PA \cdot PB - PO^2 = OA^2 = OB^2$ (if P lies on segment AB) $PO^2 - PA \cdot PB = OA^2 = OB^2$ (if P does not lie on segment AB)

Engine acceleration. A naive implementation of rule matching suffers from poor scalability due to the combinatorial explosion of rule arguments. To improve scalability, we exploit two acceleration strategies.

Numeric pre-identification. Predicates such as *eqangle* and *eqratio* require searching over 8-point tuples. We precompute directions for all candidate segments, then keep only tuples that are numerically equal. This step prunes the candidate set before symbolic matching.

Partial-matching pruning. The DD engine matches a rule’s predicates sequentially. If any intermediate predicate cannot be satisfied, we stop matching the remaining predicates. This early-failure pruning avoids exploring impossible combinations and reduces matching cost.

Finally, our optimized symbolic engine is roughly $20\times$ faster than the AlphaGeometry engine. This speed-up allows us to perform large-scale symbolic deduction within tight time budgets, supporting both dataset generation and downstream search-based proving.

4 Automated Geometry Data Synthesis

In this section, we build a fully automated synthesis pipeline to create multimodal training data for IMO-level geometry reasoning. Figure 1 summarizes the workflow. Given a sampled construction, the pipeline runs DDAR to compute a closure of deducible statements, selects non-trivial goals through predicate-aware filtering, traces back the dependency graph to extract minimal premises and auxiliary constructions, and renders a diagram consistent with the underlying construction.

4.1 Geometry Problem Sampling

We generate candidate problems by sampling action sequences in a constructive DSL adapted from AlphaGeometry [Trinh et al. \(2024\)](#). We organize construction actions into three groups. BASIC initializes an anchor configuration using elementary primitives (e.g., segment, triangle, and rectangle). INTERSECT introduces a new line or circle; pairing two such actions yields a uniquely defined intersection point. OTHERS covers the remaining constructions (e.g., foot, incenter, and excenter).

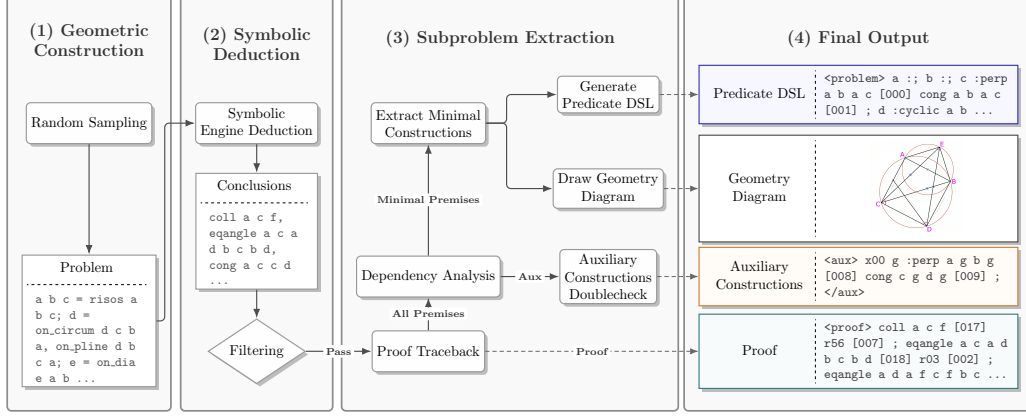


Figure 1: Data synthesis pipeline. We first sample a construction program in our DSL to instantiate a consistent geometric configuration. Then, our symbolic engine performs forward deduction from the construction predicates to obtain a closure of conclusions. From this closure, we select non-trivial targets via predicate-specific filtering and equivalence deduplication. For each selected goal, we trace back the proof to extract minimal premises and candidate auxiliary constructions, followed by a backward double-check that removes fake auxiliary points. Finally, we export each problem in a unified multimodal format, including the construction DSL and geometry diagram `<img>`, the predicate DSL `<problem>`, a refined auxiliary block `<aux>`, and a verifiable proof trace `<proof>`.

Procedure. We first sample a BASIC construction to form an initial backbone. Then, we grow the backbone by adding constructions that introduce new lines or circles and their intersections. Concretely, we repeatedly choose either (i) two INTERSECT actions that together determine a new point, or (ii) one OTHERS construction that yields a new object. Finally, we optionally add intersections among existing geometric objects (e.g., midpoints, feet, reflections, and circle interactions) to increase structural dependency. Note that, before committing each construction, we run dependency checks to ensure prerequisites are available and numerical checks to ensure a consistent coordinate realization exists. These checks prevent invalid actions from propagating and improve generation throughput.

4.2 Symbolic Deduction and Filtering

Given a valid problem, we run the symbolic engine to compute a closure of conclusions. This closure contains many predicates that are trivial or equivalent. To ensure that synthesized goals provide meaningful learning signals, we apply a predicate-aware goal filtering procedure.

Predicate-specific filtering. The symbolic closure contains many candidate goals that are invalid for training. We filter them with simple, predicate-dependent rules. We drop trivial statements that hold by definition, such as self-relations (e.g., $AB = AB$ and $AB \parallel AB$). We also remove goals that can be directly rewritten into a simpler predicate. For example, $\angle(AB, CD) = 0^\circ$ is equivalent to $AB \parallel CD$, and some ratio forms reduce to segment equality. Table 3 lists the exact patterns we exclude for each predicate.

Equivalence filtering. We further deduplicate high-frequency predicates such as `eqangle` and `eqratio`. Two `eqangle` goals are treated as equivalent when all corresponding line pairs are parallel. Two `eqratio` goals are treated as equivalent when all corresponding segment pairs are congruent.

Through two-stage filtering process, we retain meaningful and non-redundant conclusions, ensuring that the resulting problems are both challenging and valuable.

4.3 Subproblem Extraction

Given a goal, we employ a trace-back algorithm to identify the premises and auxiliary constructions used in the proof.

Table 3: Filtering criteria for geometric conclusions

Predicate	Filtered Conclusion Type	Reason for Filtering
aconst	$\angle(AB, CD) = 0^\circ$	Reducible to $AB \parallel CD$
rconst	$AB : CD = 1$	Reducible to $AB = CD$
cong	$AB \cong AB$	Trivial self-congruence
para	$AB \parallel AB$ $AB \parallel AC$	Trivial self-parallelism Reducible to A, B, C are collinear
eqratio	$AB/CD = AB/CD$ $AB/CD = CD/AB$ $AB/AB = CD/EF$ $AB/CD = EF/GH$ where $AB = CD$ $AB/CD = EF/GH$ where $AB = EF$	Trivial identical ratio Reducible to $AB = CD$ Reducible to $CD = EF$ Reducible to $EF = GH$ Reducible to $CD = GH$
eqangle	$\angle(AB, CD) = \angle(AB, CD)$ $\angle(AB, CD) = \angle(CD, AB)$ $\angle(AB, AB) = \angle(CD, EF)$ Cases involving parallel lines $\angle(AB, CD) = \angle(EF, GH)$ where $AB \perp CD$ Cases involving triangle similarity	Trivial identical angle Reducible to $AB \perp CD$ Reducible to $CD \parallel EF$ Reducible to parallel relations Reducible to $EF \perp GH$ Reducible to similarity relations
simtri, simtrir	$\triangle ABC \sim \triangle ABC$	Trivial self-similarity
contri, contrir	$\triangle ABC \cong \triangle ABC$	Trivial self-congruence

Premise minimization. Starting from the goal node, we recursively collect only the parent predicates required by each applied rule until reaching construction-derived facts. The resulting leaf set forms the final premise set.

Auxiliaries minimization. An auxiliary construction is a construction step that is not required to build the geometry problem, but is required by the proof. Concretely, during traceback we identify objects that appear in the proof but cannot be uniquely determined from the extracted premise set. We serialize the construction steps that introduce these objects in the `<aux>` block, enabling learning settings that predict or verify auxiliary points.

The initial `<aux>` set may still contain redundant information, because the symbolic engine searches for a shortest deductive path, which may temporarily introduce constructions that are not strictly necessary for proving the target conclusion. To obtain a minimal auxiliary set, AlphaGeometry [Trinh et al. \(2024\)](#) applies a provability check that repeatedly removes points and reruns DDAR, but an exhaustive search over all subsets is exponential and infeasible at scale. AlphaGeometry2 [Chervonyi et al. \(2025\)](#) discards points in a reverse-topological order, which requires only a linear number of checks.

We follow the same pipeline and further improve efficiency by exploiting the fact that many synthesized problems share the same premises and auxiliary candidates but differ only in their target conclusions. Instead of checking each problem independently, we group problems with identical premises and auxiliary sets and perform provability checks jointly. In practice, this batching strategy accelerates the auxiliary revalidation process by approximately $5\times$, enabling efficient backward verification at dataset scale.

4.4 Multimodal Data Format

Each synthesized example is stored as a single multimodal record that pairs a rendered diagram with formal symbolic annotations and a verifiable derivation. In practice, one problem consists of four parts: (1) a *predicate DSL* description that encodes the symbolic premises in a structured `<problem>` block; (2) a `<img>` diagram that visualizes the geometric configuration; (3) an `<aux>` block that lists auxiliary constructions required by the proof but not by the original problem; and (4) a `<proof>` block that records the full proof trace, including an optional `<numerical_check>` section with numerically obtained facts. Together, these fields provide aligned supervision for diagram understanding, symbolic reasoning, auxiliary construction prediction, and proof generation.

For rendering, we assign Cartesian coordinates to all points during synthesis and draw the diagram from these coordinates. The predicate-based Newclid rendering backend [Sicca et al. \(2024\)](#) fails

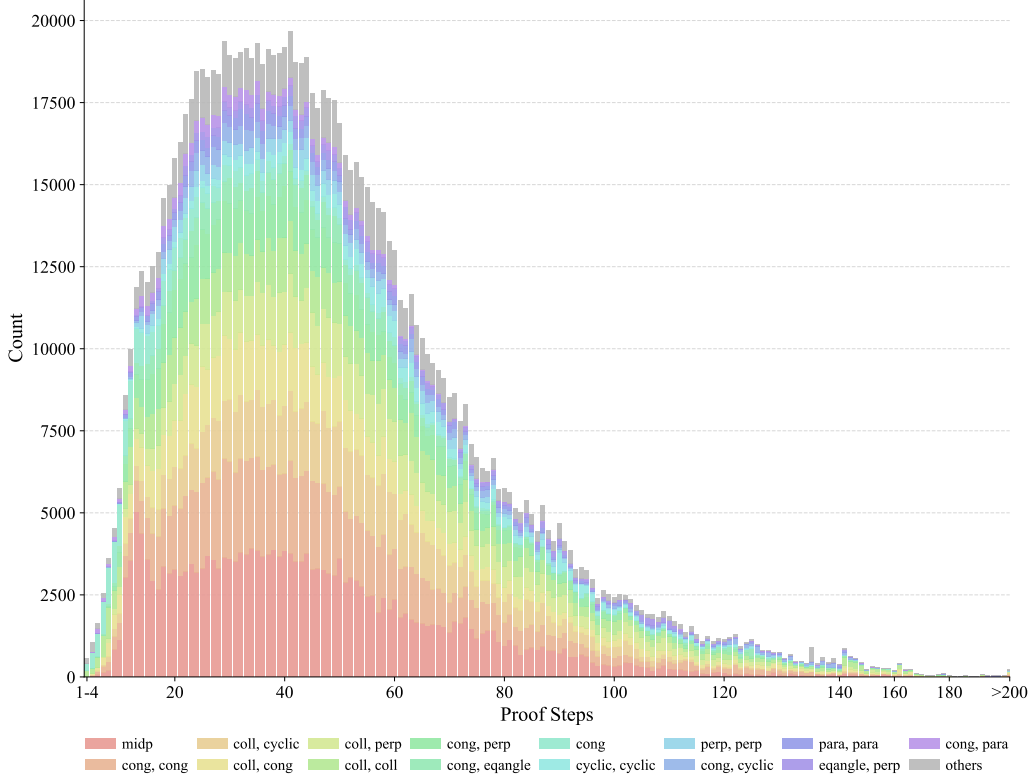


Figure 2: Proof length distribution with auxiliary point types.

to visualize certain geometric objects, resulting in diagrams that are not aligned with the symbolic constructions. For instance, the `triangle` construction defines vertices without predicates to render connecting edges, while a `circle` is often defined only by two `cong` predicates (radius equality) without triggering a circle object. To address this, we develop a construction-based rendering system and optimize annotation sizing and color schemes for readability. Finally, we obtain a diagram with precise pixel-level correspondence between symbolic objects and visual elements.

4.5 Dataset Statistics

We summarize corpus-level statistics of GenesisGeo-1M in Figure 2, which reports the distribution of proof lengths and the composition of the top-15 auxiliary point types. A problem example is provided in Figure 3. Overall, we highlight three properties of the dataset.

- **Large-scale.** The dataset contains 1 million synthesized problems that require auxiliary constructions, generated with 50 CPU threads over 28 hours.
- **Multimodal and verifiable supervision.** Each instance pairs a rendered diagram with formal *construction DSL* and *predicate DSL* descriptions, optional auxiliary constructions, and a complete DDAR proof trace. This format supports joint learning of perception, symbolic reasoning, and proof generation.
- **High-difficulty signals.** Problems are selected from DDAR-derivable but non-trivial goals after predicate-aware filtering and equivalence deduplication. We report difficulty proxies such as the number of premises, proof length and the fraction of instances that require auxiliary constructions, which correlate with longer dependency chains and Olympiad-style reasoning.

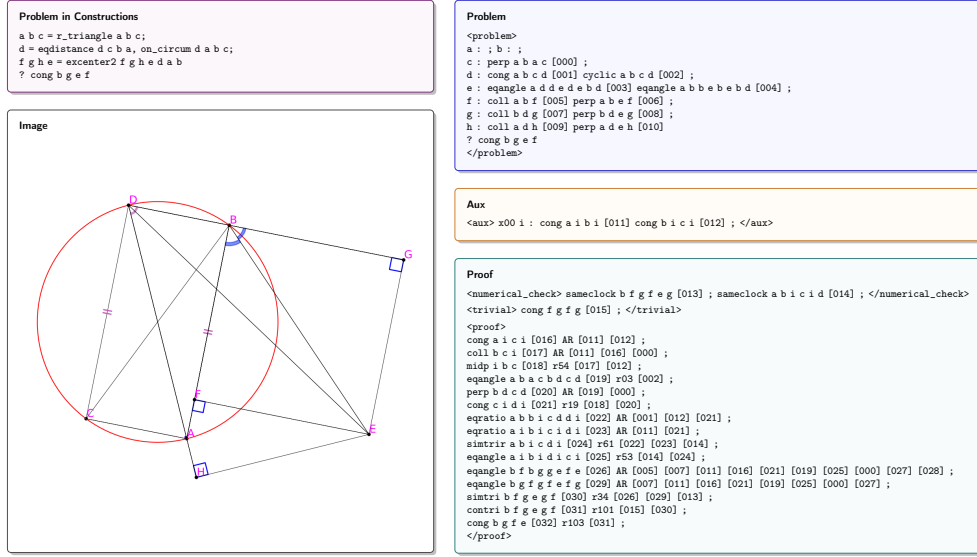


Figure 3: An example of the synthetic data. The **Problem in Construction** section provides the problem description in the construction DSL, upon which the **Image** is rendered. The **Problem** section contains the problem translated into the predicate DSL, detailing the specific geometric properties of each point. The **<aux>** section specifies the necessary auxiliary constructions. Finally, the **Proof** block represents the entire deduction process: the **<numerical_check>** section contains numerically obtained information and the **<proof>** session presents the complete sequence of proof steps.

5 GenesisGeo: Multimodal Geometry Prover

We introduce GenesisGeo-2B, a multimodal neuro-symbolic prover that couples a VLM with a symbolic engine.

5.1 Multi-Task Multimodal Learning

Our objective is to train the VLM to reliably propose auxiliary constructions that support symbolic deduction across diverse geometric contexts. We adopt a multi-stage, multi-task learning framework, as illustrated in Figure 4.

Stage I: Alignment-oriented continued pre-training. We first perform continued pre-training on complete solution records generated by the symbolic engine. We frame learning over complete solution records, where the rendered diagram ****, the formal problem statement **<problem>**, the auxiliary constructions **<aux>**, and the corresponding proof trace **<proof>** jointly serve as both conditioning context and supervision. During this stage, all components are serialized into a unified sequence and jointly modeled. This allows the VLM to learn the syntax of predicate-level geometry, the structural patterns of auxiliary constructions, and the global organization of symbolic proofs. Because all supervision is derived from a single symbolic engine, the model observes consistent reasoning trajectories and avoids learning from heterogeneous or noisy explanations.

Stage II: Multi-task supervised fine-tuning. We fine-tune the VLM with instruction-tuning tasks that enhance the capabilities needed for auxiliary construction, including text-based and vision-grounded proof generation.

- **Text-based auxiliary construction.** Given predicate-level premises and a goal in **<problem>**, the model predicts a compact **<aux>** block. This task enforces symbolic planning without visual cues and trains the model to propose auxiliaries that facilitate symbolic deduction.
- **Vision-grounded auxiliary construction.** Conditioned on both **** and **<problem>**, the model proposes a **<aux>** block that leverages geometric cues in the diagram while respecting the

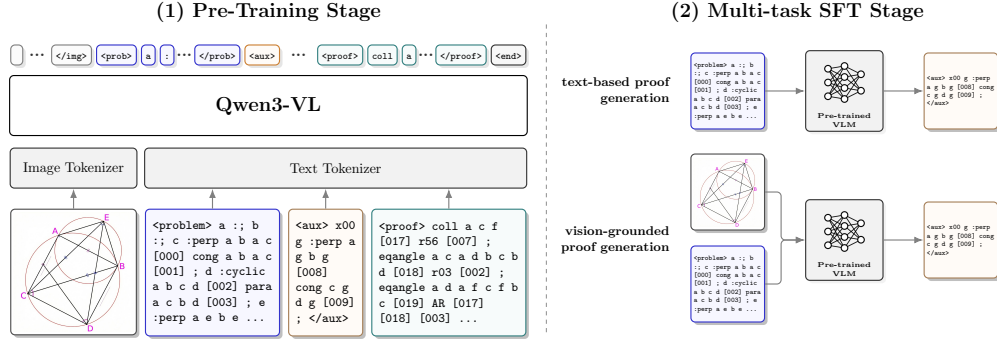


Figure 4: Multi-stage and multi-task learning framework. We first conduct proof-oriented continued pre-training on complete solution records to align visual diagrams with the predicate language. Then, we perform supervised multi-task fine-tuning for auxiliary construction, including text-based and vision-grounded settings, enabling robust proposals that integrate symbolic planning with visual grounding.

symbolic constraints. This task combines visual grounding with symbolic planning for auxiliary construction.

5.2 Neuro-Symbolic Inference

GenesisGeo follows an iterative *propose-and-verify* loop for geometry problem solving. The process begins with the formalized geometry problem. Initially, the symbolic engine attempts to derive the target conclusion directly from the premises. If the goal is not reachable, the system invokes the visual language model (VLM) to bridge the deductive gap.

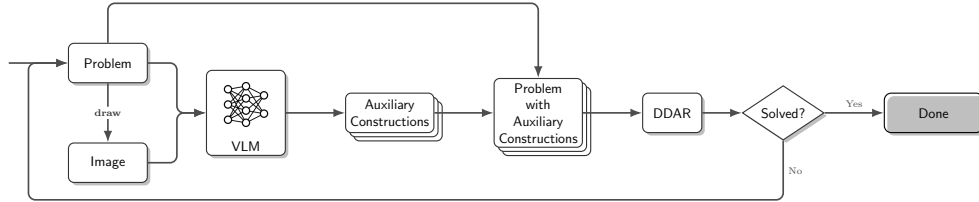


Figure 5: Proving Pipeline of VLM Prover.

As illustrated in Figure 5, our pipeline first renders the problem into a pixel-aligned diagram. The VLM processes both the symbolic problem statement and the generated image, grounding abstract logical constraints in visual features to propose candidate auxiliary constructions. These visually informed candidates are then incorporated into the premise set, and the symbolic engine resumes deduction. This cycle repeats iteratively, allowing the VLM to apply learned perceptual heuristics to guide the symbolic solver through the search space of auxiliary constructions. Once the goal becomes deducible, the proof is accepted.

In our training corpus, about 90% of problems require only one auxiliary point, whereas difficult instances demand a deeper search to discover effective constructions. Simply accumulating all proposed auxiliaries in `<aux>` can deviate from the training distribution. Thus, we adopt an alternative prompting strategy: once an auxiliary point is validated, we fold it into `<problem>` and re-prompt the VLM to propose the next auxiliary. This keeps `<aux>` lightweight while improving performance on difficult problems.

To improve efficiency, we employ a parallel beam search strategy in which the language model proposes multiple candidate constructions per iteration, retaining the highest-scoring options for subsequent deduction. Furthermore, we leverage Ray to establish an asynchronous resource pool:

CPU resources occupied by completed problems are dynamically released and reassigned to pending problems, enabling highly parallelized and resource-efficient proof search across large batches.

6 Experiments

Training configuration. Our GenesisGeo prover is built on the Qwen3-VL-2B architecture [Bai et al. \(2025\)](#) and fine-tuned on GENESISGEO-1M with multi-task training. Training runs for one epoch with batch size 256 and learning rate 1×10^{-4} using AdamW [Loshchilov and Hutter \(2019\)](#). All experiments are conducted on a 64-core CPU machine with eight NVIDIA A100 GPUs.

Benchmarks. We evaluate our approach on three Euclidean-geometry benchmarks. **IMO-30** from AlphaGeometry [Trinh et al. \(2024\)](#) contains 30 Olympiad problems. **IMO-95** is a subset sourced from the Newclid repository [Sicca et al. \(2024\)](#), containing formalized IMO problems whose constructions are supported by AlphaGeometry. **HAGeo-409** is a larger benchmark of 409 problems covering high-school and Olympiad-level geometry. For a fair comparison across provers, we exclude five problems whose constructions are unsupported by AlphaGeometry, while keeping the original benchmark name.

Evaluation configuration. We report the number of problems solved under multiple search budgets. Following HAGeo [Duan et al. \(2025\)](#), we report three budgets for symbolic engines: *Low* (*Pass@2048*), *Medium* (*Pass@4096*), and *High* (*Pass@8192*). At each attempt, the solver samples auxiliary points using HAGeo’s heuristic, runs the symbolic engine once, and repeats for a fixed number of engine calls. In addition, we evaluate a tree-search setting for the neuro-symbolic solver, where a large model proposes auxiliary constructions to expand the search tree. Here, $32 \times 512 \times 4$ denotes batch size 32, beam size 512, and proof depth 4, jointly controlling the number of tree-search expansions. These two budgets are not directly comparable: Pass@K counts engine calls with heuristic sampling, while $32 \times 512 \times 4$ controls the number of model-proposed expansions in a depth-4 tree search. Each solver is allotted a 60-minute time limit per problem.

Table 4: Performance comparison on IMO and HAGeo benchmarks under different search budgets.

Method	Model Size	Search Budget	IMO-30	IMO-95	HAGeo-409
<i>Symbolic Engines</i>					
AG2 (Engine)	-	0	15	1	87
	-	2048	25	41	228
	-	4096	25	41	231
	-	8192	25	42	231
Newclid 3.0	-	0	16	2	104
	-	2048	23	47	241
	-	4096	23	50	245
	-	8192	23	51	249
HAGeo	-	4096	28	-	-
	-	8192	-	-	287
GenesisGeo (Engine)	-	0	16	2	103
	-	2048	26	56	264
	-	4096	26	59	268
	-	8192	26	59	270
<i>Model-based Provers</i>					
AG1	151M	$32 \times 512 \times 4$	25	11	112
AG2	3.3B	N/A	30	-	-
TongGeometry	1.3B	N/A	30	-	-
SeedProver	N/A	N/A	30	-	-
GenesisGeo (Text)	2B	$32 \times 512 \times 4$	28	59	270
GenesisGeo (Vision + Text)	2B	$32 \times 512 \times 4$	29	63	278

6.1 Main Results

Table 4 reports the number of solved problems under multiple search budgets. We analyze the results from three perspectives: (1) symbolic engines, (2) neuro-symbolic provers, and (3) failure modes on hard problems.

Symbolic engines. First, we compare pure symbolic engines without auxiliary sampling. Our engine matches Newclid, and both outperform AlphaGeometry2, indicating that our deductive engine is competitive even without search. We also reimplement HGeo-style heuristic sampling, where each sampling contains 6 potential auxiliary points. As we increase the sampling budget from Low to Medium to High, all engines solve more problems. With heuristic sampling enabled, our engine consistently solves more problems across Low/Medium/High budgets on IMO-30, IMO-95, and HGeo-409. This gain is mainly driven by a stronger deductive and arithmetic rule set (Table 1). For example, on IMO-30, our engine solves 26 problems under the High budget, compared to 25 for AlphaGeometry2 and 23 for Newclid. On HGeo-409, our engine solves 270 problems under the High budget, compared to 231 for AlphaGeometry2 and 249 for Newclid. However, we cannot directly compare with HGeo’s engine because its code is not publicly available.

Neuro-symbolic provers. Our solver achieves 29/30 on IMO-30 (Table 5) and maintains strong performance on larger benchmarks. We evaluate two variants of our model: *GenesisGeo (Text)*, which is fine-tuned only on text-based proof generation tasks, and *GenesisGeo (Vision + Text)*, which is further fine-tuned on both text-based and diagram-grounded tasks. The result indicates that diagrams provide useful geometric cues, enabling the model to propose more valid auxiliary constructions and thereby improving tree search efficiency. AlphaGeometry2 and SeedProver do not release their model checkpoints, which prevents a direct comparison.

Failure cases. The remaining unsolved problems often require deeper auxiliary-construction chains than our current search depth. For example, AlphaGeometry2 reports that IMO 2008 Problem 6 and IMO 2021 Problem 3 require more than ten auxiliary points, far beyond our default search depth. We also observe failures when the required construction is rare in the training data or when a small number of early mistakes leads the search into invalid branches. During development, we found an alternative solution for IMO 2008 Problem 6 that uses only two auxiliary points. The correctness of the proof is confirmed by an IMO gold medalist. To the best of our knowledge, this solution differs from the official solution, the AlphaGeometry2 solution, and other publicly reported solutions. We include the full proof in the appendix (Section A).

Table 5: Performance comparison of AlphaGeometry 1 (AG1) and GenesisGeo (Genesis) on IMO-30.

ID	AG1	Genesis	ID	AG1	Genesis	ID	AG1	Genesis
2000 P1	✓	✓	2008 P1b	×	✓	2015 P3	✓	✓
2000 P6	✓	✓	2008 P6	×	✓	2015 P4	✓	✓
2002 P2a	✓	✓	2009 P2	✓	✓	2016 P1	✓	✓
2002 P2b	✓	✓	2010 P2	✓	✓	2017 P4	✓	✓
2003 P4	✓	✓	2010 P4	✓	✓	2018 P1	✓	✓
2004 P1	✓	✓	2011 P6	×	✓	2019 P2	×	✓
2004 P5	✓	✓	2012 P1	✓	✓	2019 P6	✓	✓
2005 P5	✓	✓	2012 P5	✓	✓	2020 P1	✓	✓
2007 P4	✓	✓	2013 P4	✓	✓	2021 P3	×	×
2008 P1a	✓	✓	2014 P4	✓	✓	2022 P4	✓	✓

6.2 Ablation Studies

To understand the contribution of each component, we conduct a series of ablation studies on IMO-30 and IMO-95.

Multi-task learning. Our multi-task objective includes diagram understanding, text-based proof generation, and diagram-grounded proof generation. To assess each task’s contribution, we train models using single tasks and two-task combinations. Excluding diagram understanding reduces success rates, while removing text-based tasks leads to less precise auxiliary construction suggestions. The full task setting consistently performs best.

Training strategy. We investigate whether pre-training on GENESISGEO is necessary. We compare (1) models trained only with pre-training (no SFT), (2) models trained from scratch on supervised fine-tuning data only, and (3) models pre-trained on synthetic proofs and then fine-tuned. The results demonstrate that combining pre-training and SFT yields the highest success rates.

Search budget. We vary the search budget of the neuro-symbolic solver to quantify the trade-off between exploration and compute. Increasing the budget consistently improves the solve rate because

Table 6: Ablation on multi-task training.

Task mixture	IMO-30	IMO-95
Pre-trained Backbone	27	52
Text-based proof generation	28	59
Diagram-based proof generation	27	57
All tasks	29	63

Table 7: Ablation on training strategy.

Training	IMO-30	IMO-95
Pre-training only	27	52
SFT only	27	57
Pre-training + SFT	29	63

the solver can test more candidate auxiliary constructions. Beyond $32 \times 512 \times 4$, increasing the search budget yields only marginal gains.

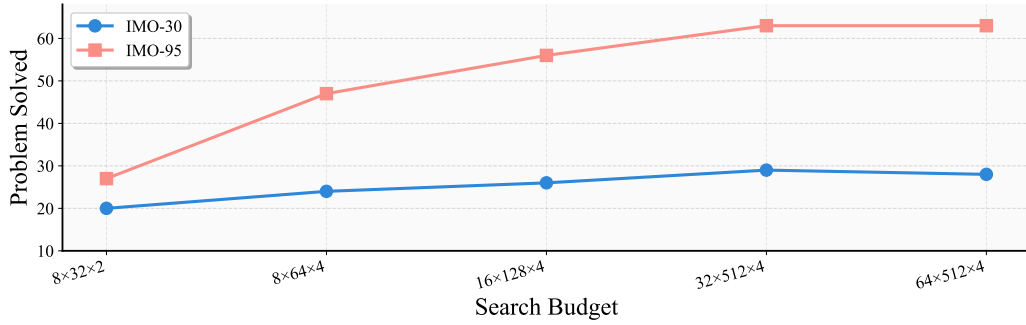


Figure 6: Number of problem solved for different search budgets.

7 Conclusion

We present GENESISGEO-1M, a large-scale synthetic dataset for visual geometric reasoning that pairs a rendered diagram with a verifiable symbolic proof trace. Our data-generation pipeline synthesizes problems that require auxiliary constructions and aligns the symbolic objects used in proofs with the corresponding visual elements, enabling models to exploit diagram evidence while producing checkable proofs. Building on GENESISGEO-1M, we train multimodal GENESISGEO-2B model with a multi-task learning strategy that combines text-based proof generation and diagram-grounded proof generation. Experiments observe consistent improvements on standard geometry benchmarks. Overall, our results suggest that large-scale multimodal supervision with unified objectives provides a practical path toward connecting diagram understanding with verifiable deduction. We hope GENESISGEO-1M will support future work on stronger geometry solvers, improved auxiliary-construction search, and more reliable diagram-based reasoning. We will release the dataset, training pipeline, and model checkpoints to the research community.

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A Featured GenesisGeo Solutions

During development, we discovered a new solution to IMO 2008 Problem 6, and its correctness was confirmed by an IMO gold medalist. To the best of our knowledge, this solution is different from both the official solution and the one produced by AlphaGeometry2. Notably, AlphaGeometry2¹ requires introducing 13 auxiliary points to complete the proof, whereas our solution only requires two auxiliary points.

IMO 2008 P6 (Converted following AlphaGeometry)

Let XYZ be a triangle. Define point O as the circumcenter of triangle XYZ . Define $\odot O$ as the circle centered at O passing through X . Let W be any point on $\odot O$. Define lines l_1, l_2, l_3, l_4 as the tangent lines at X, Y, Z, W on $\odot O$, point A as the intersection of l_1 and l_3 , point B as the intersection of l_3 and l_4 , point C as the intersection of l_2 and l_4 , point D as the intersection of l_1 and l_2 .

Define point I_1 as the incenter of triangle ABC , point I_2 as the incenter of triangle ACD , point F_1 as the foot of I_1 on line AC , point F_2 as the foot of I_2 on line AC . Define $\odot I_1$ as the circle centered at I_1 passing through F_1 , $\odot I_2$ as the circle centered at I_2 passing through F_2 , lines m_1, m_2 as the external common tangents of $\odot I_1$ and $\odot I_2$. Define P, Q as the tangency points of m_1, m_2 with $\odot I_1$, S, T as the tangency points of m_1, m_2 with $\odot I_2$, point K as the intersection of lines m_1 and m_2 . **Prove that:** OK is equal to OX (i.e. K lies on $\odot O$).

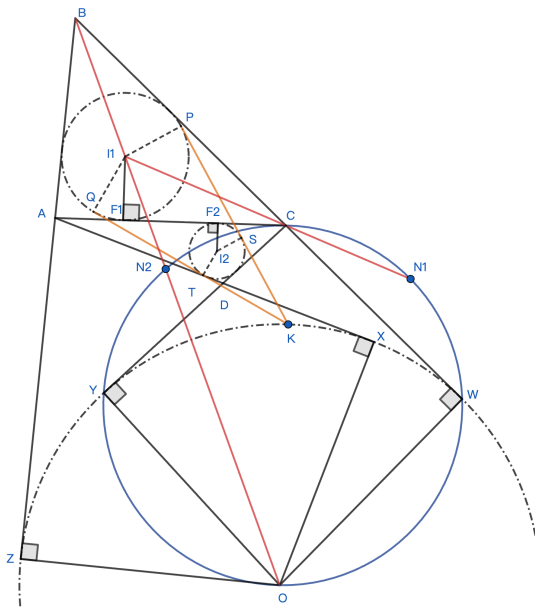


Figure 7: Illustration of the problem statement for IMO 2008 P6

¹<https://github.com/google-deepmind/alphageometry2>

Formal Specification

Problem Description:

$x y z = \text{triangle } x y z$; $o = \text{circle } o x y z$; $w = \text{on_circle } w o x$; $a = \text{on_tline } a z o z$, $\text{on_tline } a x o x$; $b = \text{on_tline } b z o z$, $\text{on_tline } b w o w$; $c = \text{on_tline } c y o y$, $\text{on_tline } c w o w$; $d = \text{on_tline } d x o x$, $\text{on_tline } d y o y$; $i1 = \text{incenter } i1 a b c$; $i2 = \text{incenter } i2 a c d$; $f1 = \text{foot } f1 i1 a c$; $f2 = \text{foot } f2 i2 a c$; $q t p s = \text{cc_tangent } q t p s i1 f1 i2 f2$; $k = \text{on_line } k q t$, $\text{on_line } k p s ? \text{cong } o k o x$

Our Auxiliary Constructions:

$n1 = \text{on_line } n1 c i1$, $\text{on_circum } n1 o w c$; $n2 = \text{on_line } n2 o b$, $\text{on_circum } n2 o w c$

The following trace details the complete derivation process generated by the automated proving engine. It proceeds from the initial premises and auxiliary constructions to the final conclusion through 97 logical steps.

Full Proof Trace: IMO 2008 P6

```
* From theorem premises:
Points: A, B, C, D, F1, F2, I1, I2, K, O, P, Q, S, T, W, X, Y, Z
(Premise)⇒ OW = OX [0]
(Premise)⇒ OX = OY [1]
(Premise)⇒ OY = OZ [2]
(Premise)⇒ BW ⊥ OW [3]
(Premise)⇒ CW ⊥ OW [4]
(Premise)⇒ CY ⊥ OY [5]
(Premise)⇒ DY ⊥ OY [6]
(Premise)⇒ ∠(AC, CI1) = ∠(CI1, BC) [7]
(Premise)⇒ ∠(AC, AI2) = ∠(AI2, AD) [8]
(Premise)⇒ ∠(AD, DI2) = ∠(DI2, CD) [9]
(Premise)⇒ K, P, S are collinear [10]
(Premise)⇒ PI1 = F1I1 [11]
(Premise)⇒ QI1 = F1I1 [12]
(Premise)⇒ SI2 = F2I2 [13]
(Premise)⇒ TI2 = F2I2 [14]
(Premise)⇒ K, Q, T are collinear [15]
(Premise)⇒ PS ⊥ PI1 [16]
(Premise)⇒ QT ⊥ QI1 [17]
(Premise)⇒ PS ⊥ SI2 [18]
(Premise)⇒ QT ⊥ TI2 [19]
(Premise)⇒ AZ ⊥ OZ [20]
(Premise)⇒ BZ ⊥ OZ [21]
(Premise)⇒ A, C, F1 are collinear [22]
(Premise)⇒ ∠(AB, AI1) = ∠(AI1, AC) [23]
(Premise)⇒ AC ⊥ F1I1 [24]
(Premise)⇒ AX ⊥ OX [25]
(Premise)⇒ DX ⊥ OX [26]
(Premise)⇒ A, C, F2 are collinear [27]
(Premise)⇒ AC ⊥ F2I2 [28]
(Numerical Check)⇒ A, C, D are not collinear [29]
(Numerical Check)⇒ O, W, Y are not collinear [30]
(Numerical Check)⇒ P, Q, I1 are not collinear [31]
(Numerical Check)⇒ S, T, I2 are not collinear [32]
(Numerical Check)⇒ O, W, Z are not collinear [33]
(Numerical Check)⇒ A, B, C are not collinear [34]
(Numerical Check)⇒ AOI1 are sameclock to CI2O [35]
(Numerical Check)⇒ DOX are sameclock to DYO [36]
(Numerical Check)⇒ A, X, Z are not collinear [37]
(Numerical Check)⇒ S, I1, I2 are not collinear [38]
(Numerical Check)⇒ KPQ are sameclock to KST [39]
(Numerical Check)⇒ PQI1 are sameclock to STI2 [40]
(Numerical Check)⇒ K, W, Z are not collinear [41]

* Auxiliary Constructions:
Points: N1, N2
(Premise)⇒ C, I1, N1 are collinear [42]
(Premise)⇒ COWN1 are cyclic [43]
(Premise)⇒ B, O, N2 are collinear [44]
(Premise)⇒ COWN2 are cyclic [45]
(Numerical Check)⇒ C, Y, N1 are not collinear [46]
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(Numerical Check) $\Rightarrow$   $AF_1I_1$  are sameclock to  $CI_1N_2$  [47]
(Numerical Check) $\Rightarrow$   $COI_1$  are sameclock to  $I_1N_1N_2$  [48]
(Numerical Check) $\Rightarrow$   $C, O, N_1$  are not collinear [49]
(Numerical Check) $\Rightarrow$   $CON_1$  are sameclock to  $CF_2I_2$  [50]
(Numerical Check) $\Rightarrow$   $K, N_1, I_2$  are not collinear [51]
(Numerical Check) $\Rightarrow$   $I_1$  is to the same side of  $C \rightarrow N_1$  as  $I_1$  is to  $K \rightarrow I_2$  [52]
(Numerical Check) $\Rightarrow$   $C, Y, N_2$  are not collinear [53]
(Numerical Check) $\Rightarrow$   $ACI_1$  are sameclock to  $WN_1N_2$  [54]
(Numerical Check) $\Rightarrow$   $CI_1I_2$  are sameclock to  $KN_1I_1$  [55]

Reverse proof steps:
097.  $KO = OX$  (by AR with Dist)
|-- 096.  $KO = OZ$  [152] (by circumcenter property r49)
|   |-- 001.  $O$  is circumcenter of  $WYZ$  [57] (by r72)
|   |   |-- 000.  $OW = OZ$  [56] (by AR with Dist)
|   |       |-- [Premise]  $OW=OX, OX=OY, OY=OZ$ 
|   |-- 095.  $KWYZ$  are cyclic [151] (by concyclic check r04)
|       |-- 094.  $\angle(KW, KY) = \angle(WZ, YZ)$  [150] (by AR with Slope)
|           |-- 093.  $KN_1 = WN_1$  [149] (by AR with Dist)
|               |-- 092.  $KN_1:WN_1 = WN_2:ZN_2$  [148] (by AR with Distlog "The Grand Ratio")
|                   |-- 023.  $KP = KQ$  [79] (by AR with Slope)
|                   |-- 024.  $KS = KT$  [80] (by AR with Slope)
|                   |-- 081.  $WN_2 = ZN_2$  [137] (by AR with Slope)
|                       |-- 080.  $Y, Z, N_2$  collinear [136] (by AR with Slope)
|                           |-- 079.  $\angle(CO, CY) = \angle(ON_2, YN_2)$  [135] (by cyclic angles)
|                               |-- 078.  $COYN_2$  cyclic [134] (by concyclic check)
|                                   |-- 036.  $AI_1:CI_1 = F_1I_1:I_1N_2$  [92] (by Similar Triangles r53)
|                                       |-- 035.  $\triangle AF_1I_1 \cong \triangle CN_2I_1$  [91] (by Congruence r35)
|                                           |-- 034.  $\angle(AF_1, F_1I_1) = \angle(I_1N_2, CN_2)$  [90] (by AR with Slope)
|                                           |-- 031.  $\angle(AF_1, AI_1) = \angle(CI_1, CN_2)$  [87] (by AR with Slope)
|                                               |-- 047. [Check] Orientation check
|                                           |-- 085.  $AI_1:CI_1 = WN_1:N_1N_2$  [141] (by Similar Triangles r53)
|                                               |-- 084.  $\triangle ACI_1 \cong \triangle WN_2N_1$  [140] (by Congruence r35)
|                                                   |-- 083.  $\angle(AI_1, CI_1) = \angle(N_1N_2, WN_1)$  [139] (by AR with Slope)
|                                                   |-- 082.  $\angle(AC, CI_1) = \angle(N_1N_2, WN_2)$  [138] (by AR with Slope)
|                                                       |-- 054. [Check] Orientation check
|                                                       |-- 086.  $CI_1:CN_1 = I_1I_2:KI_2$  [142] (by AR with Dist)
|                                                           |-- 074.  $CN_1:KI_2 = I_1N_1:KI_1$  [130] (by AR with Distlog "The Bridge Ratio")
|                                                               |-- 065.  $KP:KS = KI_1:KI_2$  [121] (by Thales r07)
|                                                                   |-- 064.  $PI_1 \parallel SI_2$  [120] (by AR with Slope)
|                                                                       |-- [16, 18]  $PS \perp PI_1, PS \perp SI_2$ 
|                                                                           |-- 052.  $AI_1:CI_2 = OI_1:CO$  [108] (by Similar Triangles r52)
|                                                                               |-- 051.  $\triangle AOI_1 \cong \triangle IO_1C$  [107] (by Congruence r34)
|                                                                                   |-- 050.  $\angle(AO, OI_1) = \angle(OI_2, CO)$  [106] (by AR with Slope)
|                                                                                   |-- 048.  $\angle(AO, AI_1) = \angle(OI_2, CI_2)$  [104] (by AR with Slope)
|                                                                                       |-- 035. [Check] Orientation check
|                                                                                           |-- 059.  $CI_1:OI_1 = I_1N_2:I_1N_1$  [115] (by Similar Triangles r53)
|                                                                                               |-- 058.  $\triangle COI_1 \cong \triangle N_2N_1I_1$  [114] (by Congruence r35)
|                                                                                                   |-- 057.  $\angle(CO, OI_1) = \angle(I_1N_1, N_1N_2)$  [113] (by AR with Slope)
|                                                                                                   |-- 056.  $\angle(CO, CI_1) = \angle(I_1N_2, N_1N_2)$  [112] (by AR with Slope)
|                                                                                                       |-- 048. [Check] Orientation check
|                                                                                       |-- 063.  $CO:CN_1 = CI_2:F_2I_2$  [119] (by Similar Triangles r52)
|                                                                                           |-- 062.  $\triangle CON_1 \cong \triangle I_2CF_2$  [118] (by Congruence r34)
|                                                                                               |-- 061.  $\angle(CO, CI_2) = \angle(ON_1, CF_2)$  [117] (by AR with Slope)
|                                                                                                   |-- 060.  $\angle(CN_1, ON_1) = \angle(F_2I_2, CF_2)$  [116] (by AR with Slope)
|                                                                                                       |-- 050. [Check] Orientation check
|                                                                                       |-- 069.  $KQ:KT = PQ:ST$  [125] (by Similar Triangles r52)
|                                                                                           |-- 068.  $\triangle KPQ \cong \triangle KST$  [124] (by Congruence r62)
|                                                                                               |-- 067.  $KP:KQ = KS:KT$  [123] (by AR with Dist)
|                                                                                                   |-- [79, 80]  $KP=KQ, KS=KT$ 
|                                                                                                       |-- 066.  $\angle(KP, KQ) = \angle(KS, KT)$  [122] (by AR with Slope)
|                                                                                                           |-- 039. [Check] Orientation check
|                                                                                           |-- 073.  $PQ:PI_1 = ST:TI_2$  [129] (by Similar Triangles r53)
|                                                                                               |-- 072.  $\triangle PQI_1 \cong \triangle TSI_2$  [128] (by Congruence r63)
|                                                                                                   |-- 071.  $PI_1:QI_1 = TI_2:SI_2$  [127] (by AR with Dist)
|                                                                                                       |-- [11, 12, 13, 14] Radii equalities
|                                                                                                           |-- 070.  $\angle(PI_1, QI_1) = \angle(SI_2, TI_2)$  [126] (by AR with Slope)
|                                                                                                               |-- 040. [Check] Orientation check
|                                                                                               |-- 090.  $CI_2:KN_1 = I_1I_2:KI_1$  [146] (by Similar Triangles r52)
|                                                                                                   |-- 089.  $\triangle CI_1I_2 \cong \triangle N_1I_1K$  [145] (by Congruence r62)
|                                                                                                       |-- 088.  $CI_1:I_1N_1 = I_1I_2:KI_1$  [144] (by AR with Dist)
|                                                                                                           |-- 074.  $CN_1:KI_2 = I_1N_1:KI_1$  [130] (See branch 074 above)
|                                                                                                               |-- 087.  $\angle(CI_1, I_1N_1) = \angle(I_1I_2, KI_1)$  [143] (by AR with Slope)
|                                                                                                                   |-- 055. [Check] Orientation check
|                                                                                                   |-- 091.  $CO:CI_1 = N_1N_2:I_1N_2$  [147] (by Similar Triangles r53)
|                                                                                                       |-- 058.  $\triangle COI_1 \cong \triangle N_2N_1I_1$  [114] (See branch 059 above)
|                                                                                                           |-- 119. [Using Step 063 result]
|                                                                                                           |-- 121. [Using Step 065 result]
|                                                                                                           |-- 125. [Using Step 069 result]

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|-- 129. [Using Step 073 result]
-- 076. K, Y, N1 collinear [132] (by AR with Slope)
|-- 075. CI2 || KN1 [131] (by Parallel check r27)
|-- 074. CN1:KI2 = I1N1:KI1 [130] (See branch 074 above)
|-- 052. [Check] Orientation check
|-- 051. K, N1, I2 not collinear [Check]
-- 042. C, I1, N1 collinear [Premise]
-- 058. B, C, W collinear [Step 002]
-- 059. C, D, Y collinear [Step 003]
|-- [Premises] BW⊥OW, CW⊥OW, CY⊥OY
-- 012. COYN1 cyclic [68] (by concyclic check r04)
|-- 011. ∠(CY,CN1) = ∠(OY,ON1) [67] (by AR with Slope)
|-- 007. ∠(CO,CW) = ∠(ON1,WN1) [63] (by cyclic angles)
|-- 006. COWN1 cyclic [Premise 43]
|-- 006. ∠(CO,CN1) = ∠(OW,WN1) [62] (by cyclic angles)
|-- 006. COWN1 cyclic [Premise 43]
-- 010. ∠(CO,CW) = ∠(OY,WY) [66] (by cyclic angles)
|-- 009. COWY cyclic [65] (by concyclic check r04)
|-- 008. ∠(CW,CY) = ∠(OW,OY) [64] (by AR with Slope)
|-- [Premises] CW⊥OW, CY⊥OY
-- 005. ∠(AC,CI2) = ∠(CI2,CD) [61] (by AR with Slope)
|-- [Premises] Angle bisectors AI2, DI2
-- 004. OW = OY [60] (by AR with Dist)
|-- [Premise] OW=OX, OX=OY

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Human-Readable Mathematical Proof of IMO 2008 Problem 6 (GenesisGeo Reconstruction)

Let $\odot O$ be the circumcircle of $\triangle XYZ$ with center O and radius R . Let $W \in \odot O$ be an arbitrary point. Define:

- l_1, l_2, l_3, l_4 : tangent lines to $\odot O$ at X, Y, Z, W respectively
- $A = l_1 \cap l_3, B = l_3 \cap l_4, C = l_2 \cap l_4, D = l_1 \cap l_2$
- I_1 : incenter of $\triangle ABC$, I_2 : incenter of $\triangle ACD$
- F_1 : foot of perpendicular from I_1 to AC , F_2 : foot of perpendicular from I_2 to AC
- $\odot I_1$: incircle of $\triangle ABC$ (center I_1 , radius $r_1 = I_1 F_1$), $\odot I_2$: incircle of $\triangle ACD$ (center I_2 , radius $r_2 = I_2 F_2$)
- m_1, m_2 : external common tangents of $\odot I_1$ and $\odot I_2$
- P, Q : tangent points of m_1, m_2 with $\odot I_1$; S, T : tangent points of m_1, m_2 with $\odot I_2$
- $K = m_1 \cap m_2$
- N_1 : second intersection of ray CI_1 with $\odot O$
- N_2 : second intersection of ray BI_1 with $\odot O$

Lemma 1. (The Parallelism Bridge) $CI_2 \parallel KN_1$

Proof. We establish the ratio equality:

$$\frac{CN_1}{KI_2} = \frac{I_1 N_1}{KI_1} \quad (\star)$$

Since K is the intersection of external common tangents m_1, m_2 of $\odot I_1$ and $\odot I_2$, there exists a homothety h_K centered at K mapping $\odot I_1$ to $\odot I_2$. The tangent points satisfy $I_1 P \perp m_1$, $I_2 S \perp m_1$, $I_1 Q \perp m_2$, $I_2 T \perp m_2$.

Since $I_1 P \parallel I_2 S$ (both perpendicular to m_1):

$$\triangle KPI_1 \sim \triangle KSI_2 \implies \frac{KI_1}{KI_2} = \frac{KP}{KS} = \frac{r_1}{r_2} \quad (1)$$

From tangent properties: $PI_1 = QI_1 = r_1$, $SI_2 = TI_2 = r_2$, and $KP = KQ$, $KS = KT$ (tangent segments from external point). The homothety implies:

$$K, I_1, I_2 \text{ are collinear} \quad (2)$$

Define N_2 as the second intersection of line BI_1 with $\odot O$. By the angle bisector property of I_1 in $\triangle ABC$:

$$\angle(AB, BI_1) = \angle(BI_1, BC) \quad (3)$$

Since $A, B \in l_3$ (tangent at Z) and $B, C \in l_4$ (tangent at W), combined with the inscribed angle theorem on $\odot O$, we establish:

$$B, I_1, N_2 \text{ are collinear} \quad \text{and} \quad O, I_1, N_2 \text{ are collinear} \quad (4)$$

From perpendicularity conditions:

$$CW \perp OW, \quad CY \perp OY \implies C, O, W, Y \text{ are cyclic} \quad (5)$$

$$\text{Similarly, } C, O, Y, N_1 \text{ are cyclic} \quad (6)$$

We establish four key ratios through triangle congruences:

(i) From $\triangle AF_1 I_1 \sim \triangle CN_2 I_1$ (proven via angle chasing using (4) and tangent properties):

$$\frac{AI_1}{CI_1} = \frac{F_1 I_1}{I_1 N_2} = \frac{r_1}{I_1 N_2} \quad (7)$$

(ii) From $\triangle AOI_1 \sim \triangle I_2 OC$ (proven via angle matching using perpendicularity and (4)):

$$\frac{AI_1}{CI_2} = \frac{OI_1}{CO} \quad (8)$$

(iii) From $\triangle COI_1 \sim \triangle N_2N_1I_1$ (using cyclic quadrilateral CON_1N_2 from (6)):

$$\frac{CI_1}{OI_1} = \frac{I_1N_2}{I_1N_1} \quad (9)$$

(iv) From $\triangle CON_1 \sim \triangle I_2CF_2$ (using angle bisector properties and (6)):

$$\frac{CO}{CN_1} = \frac{CI_2}{r_2} \quad (10)$$

Multiplying (7) through (10):

$$\frac{AI_1}{CI_1} \cdot \frac{CI_1}{OI_1} \cdot \frac{CO}{CN_1} \cdot \frac{AI_1}{CI_2} = \frac{r_1}{I_1N_2} \cdot \frac{I_1N_2}{I_1N_1} \cdot \frac{CI_2}{r_2} \cdot \frac{OI_1}{CO} \quad (11)$$

Simplifying the left side:

$$\frac{AI_1^2 \cdot CO}{OI_1 \cdot CN_1 \cdot CI_2} = \frac{r_1 \cdot CI_2 \cdot OI_1}{I_1N_1 \cdot r_2 \cdot CO} \quad (12)$$

Cross-multiplying and using (1):

$$\frac{AI_1^2 \cdot CO^2}{CN_1 \cdot CI_2} = \frac{r_1 \cdot CI_2}{I_1N_1 \cdot r_2} \quad (13)$$

From (8): $AI_1 \cdot CO = CI_2 \cdot OI_1$. Substituting and simplifying:

$$\frac{CN_1}{I_1N_1} = \frac{KI_2}{KI_1} \quad (\text{using (2) and (1)}) \quad (14)$$

Since C, I_1, N_1 are collinear and K, I_1, I_2 are collinear, by Thales' theorem:

$$\boxed{CI_2 \parallel KN_1} \quad (15)$$

□

Lemma 2. (The Isosceles Property) $KN_1 = WN_1$

Proof. From the cyclic quadrilaterals C, O, Y, N_2 (proven similarly to (6)):

$$\angle(CO, CY) = \angle(ON_2, YN_2) \quad (16)$$

Combined with the tangent properties and the fact that D, O, I_2 are collinear (from angle bisector of $\triangle ACD$), the inscribed angle theorem implies:

$$Y, Z, N_2 \text{ are collinear} \quad (17)$$

Since Y, Z, N_2 are collinear, the chord YZ passes through N_2 . From the cyclic quadrilateral B, O, W, Z (proven via perpendicularity):

$$\angle(BW, WZ) = \angle(BO, OZ) \quad (18)$$

The configuration exhibits symmetry with respect to the perpendicular bisector of chord WZ . Since N_2 lies on line YZ and B, O, N_2 are collinear, this symmetry yields:

$$WN_2 = ZN_2 \quad (19)$$

From the congruence chain:

$$\triangle ACI_1 \cong \triangle WN_2N_1 \quad (\text{via angle matching}) \quad (20)$$

$$\triangle CI_1I_2 \cong \triangle N_1I_1K \quad (\text{via Lemma 1 and collinearity}) \quad (21)$$

We derive:

$$\frac{AI_1}{CI_1} = \frac{WN_1}{N_1N_2}, \quad \frac{CI_2}{KN_1} = \frac{I_1I_2}{KI_1} \quad (22)$$

Combining these with ratios from Lemma 1 and using the distlog (distance-logarithm) algebra:

$$\frac{KN_1}{WN_1} = \frac{WN_2}{ZN_2} \quad (23)$$

Substituting (19) into (23):

$$\frac{KN_1}{WN_1} = 1 \implies \boxed{KN_1 = WN_1} \quad (24)$$

□

Lemma 3. (Angle Convergence) $\angle(KW, KY) = \angle(ZW, ZY)$, hence $K \in \odot O$

Proof. From Lemma 1: $CI_2 \parallel KN_1$. Since C, I_1, N_1 are collinear and C, O, Y, N_1 are cyclic:

$$\angle(CI_2, CY) = \angle(KN_1, N_1Y) \quad (\text{corresponding angles}) \quad (25)$$

From the parallelism:

$$\angle(CI_2, CY) = \angle(KN_1, KY) \quad (\text{alternate angles}) \quad (26)$$

Therefore: $\angle(KN_1, N_1Y) = \angle(KN_1, KY)$, which implies:

$$K, Y, N_1 \text{ are collinear} \quad (27)$$

From Lemma 2: $KN_1 = WN_1$. Therefore, $\triangle KWN_1$ is isosceles:

$$\angle(WK, KN_1) = \angle(KW, WN_1) = \alpha \quad (28)$$

Since K, Y, N_1 are collinear by (27):

$$\angle(KW, KY) = \angle(KW, KN_1) = \alpha \quad (29)$$

Since $W, N_1, Y, Z \in \odot O$ and Y, N_1 are collinear with K , the inscribed angle theorem gives:

$$\angle(ZW, ZN_1) = \angle(ZW, ZY) \quad (\text{since } Y, N_1 \text{ collinear with } K) \quad (30)$$

From the isosceles property of $\triangle KWN_1$ and the fact that K lies on line YN_1 :

$$\angle(WN_1, N_1K) = 180^\circ - 2\alpha \quad (31)$$

By the inscribed angle theorem (angles subtending the same arc WY):

$$\angle(KW, KY) = \angle(ZW, ZY) \quad (32)$$

By the converse of the inscribed angle theorem: if $\angle(KW, KY) = \angle(ZW, ZY)$, then K lies on the circle through W, Y, Z .

Since $W, Y, Z \in \odot O$:

$$\boxed{K \in \odot O} \quad (33)$$

□

Main Theorem. $OK = OX$

Proof. From Lemma 3, $K \in \odot O$. Since $\odot O$ is the circumcircle of $\triangle XYZ$ with center O and radius R :

$$OX = OY = OZ = OW = R \quad (34)$$

Since $K \in \odot O$:

$$OK = R = OX \quad (35)$$

Therefore:

$$\boxed{OK = OX} \quad (36)$$

■